Dry Fruit House Operations Platform

Product plan for an in-house POS, inventory, procurement, finance and analytics stack for a 500 store dry fruit retail chain

Prepared by Akshay | Draft v1 | 31 October 2026

This book is the plan. The working demo runs on synthetic data modeled on the public catalog.

Where we start

What we know, what we assume, and the thesis for the demo

What we know

- Multi-store dry fruit and nut retailer based in Bengaluru, with stores across India and a possible Middle East presence.
- The public site lists six collections: Nuts and Dry Fruits, Dates, DFH Exclusives, Berries, Seeds and More, and Gift Boxes. It also carries bulk ordering, locations and careers pages. The page title mentions chocolates and spices.
- Stores bill on an older POS. Getting reports out of it automatically is hard, so finance reconciles by hand.
- No organized WMS. Analytics is manual. Oracle Cloud ERP is under discussion.
- Goods are imported and bought domestically from many vendors. The same SKU can come from several vendors at different costs across months.
- Round 2 audience is likely the CEO.

What we assume for the demo

- About 500 stores, 20 warehouses, 1,200 SKUs, shelf lives from a few weeks (fresh and packed) to a year.
- Returns are rare for food, so the demo has no returns flow. Wastage is modeled in full.
- Store expansion is out of scope.

The thesis

The pain looks like three problems: store data is stuck inside a POS, stock is not traceable by batch and expiry, and finance rebuilds the picture by hand every month. A full ERP rollout solves a bigger problem than the one in front of them, and it takes a long time to land.

The proposal is a lean platform they own. It has an offline-first POS for stores, an order and warehouse system with procurement and inbound logistics tracking, finance and GST reconciliation, and analytics on top. Every transaction lands in one database that the CEO and finance can query on day one.

Where Oracle still fits

General ledger, payroll and group consolidation are mature ERP territory. The platform exports clean journals and masters, so it works with Oracle, Tally or anything else later. The pitch is to run operations on a system they control now, and to connect an ERP when the business needs one.

What would change this plan

- If their POS vendor can give a nightly database export, Phase 1 can start with analytics before any POS replacement.
- If they already have a sound finance stack, the finance module shrinks to reconciliation only.

Scope map

A POS foundation plus the ten modules agreed earlier. Depth shows how much is live in the demo.

#	Module	What it solves	Demo depth
F	Store POS, offline first	Replaces dependence on the legacy POS. Bills, cash, GST invoices and stock movement captured in their own database, working without internet.	Deep
1	Control Tower	One screen for the CEO: sales, margin, stock, wastage, exceptions, store league table.	Deep
2	SKU master	Single catalog with packs, HSN, GST, shelf life, batch MRP, kits, price lists.	Deep
3	Vendors, procurement and inbound logistics	Vendor price book, landed cost, PO to shipment to GRN tracking, vendor scorecards.	Deep
4	Inventory and warehouse (OMS and WMS)	Batch and expiry stock ledger, FEFO, transfers, repacking, store indents, counts, bulk orders.	Deep
5	Wastage and shrink	Reason coded loss capture, near-expiry actions, shrink by location, SKU and vendor.	Deep
6	Demand and replenishment	Seasonal forecast, reorder suggestions, festive planning, assortment by store cluster.	Medium
7	Finance and tax reconciliation	Day-close reconciliation, GST working data, ITC matching, payables, stock valuation, journal export.	Deep
8	Customers and WhatsApp commerce	Customer 360, loyalty, segments, WhatsApp ordering, corporate gifting enquiries.	Light
9	HR basic	Org chart, onboarding, attendance, rosters, leave, birthdays, job postings.	Light
10	Platform	Roles, audit, approvals, notifications, imports, multi-entity and multi-currency.	Medium

Order of build follows the depth column. If time runs short, modules 8, 9 and 6 are shown as designed screens with seeded data and the rest of the demo stays intact.

Architecture

One modular backend, one database, a PWA for stores, and an analytics layer on the same data

Layer	Choice	Why
Store device	Progressive web app on a tablet or low-cost PC. IndexedDB for local data. Service worker for offline and background sync.	No app store, quick rollout, works on hardware they already own.
Sync	Append-only outbox on the device. Idempotency key per bill. Server acknowledges and dedupes. Pull-only catalog and price versions.	Bills never edit in place, so conflicts do not arise for sales.
Backend	Python 3.12, FastAPI, SQLAlchemy, Pydantic. Modular monolith, one module per domain above.	Fast to build, easy for a small team to own.
Database	PostgreSQL. Stock as an append-only ledger. Money in paise, quantity in grams or units.	Every rupee and every gram traceable to a document.
Jobs	Scheduled jobs for nightly rollups, expiry alerts, forecasts, reconciliation.	Keeps heavy analytics off the billing path.
Analytics	Materialized views for dashboards. DuckDB and Parquet for the large synthetic history.	Fast drilldowns without a separate warehouse in phase 1.
Frontend	React with Vite and Tailwind. Two densities: touch-friendly POS and compact back office.	One design system, two usage modes.
Hosting	Render for app, Neon or managed Postgres for data. Paid tiers for the demo week to avoid cold starts.	Free tiers sleep and a sleeping demo costs credibility.
Integration	CSV and JSON exports, REST API with keys, journal export for Oracle or Tally, GSP hook for e-invoice and e-way bill in phase 2.	Works alongside any future ERP.

Foundation: Store POS, offline first

Demo depth: Deep | Replaces the legacy POS in pilot stores

The goal is to show that a POS can be built in house, work through internet drops, and give the company direct access to its own sales data. The demo will bill, lose connectivity on stage, keep billing, and sync when the connection returns.

Billing

- Barcode scan, search and quick keys for top sellers.
- Loose items by weight: price per kg, grams entered, live line total.
- Packed items by scan. Gift boxes as kits with component deduction.
- Hold and recall bills. Line and bill discounts with reason and approval limits.
- Coupons and loyalty redemption. Customer lookup by phone.
- Split payments across cash, UPI, card.
- GST invoice with HSN, tax split and rounding. Print, WhatsApp or SMS receipt.

Control

- Cashier PIN login and roles. Price override limited to managers.
- Void and cancel with reason and approval. Full audit trail.
- Day open and close with cash count and variance.
- X and Z reports. Shift handover. Cash drops and petty cash.
- Business date logic for stores that close after midnight.

Store tasks beyond billing

- Stock lookup for the store and nearby stores.
- Receive transfers and deliveries by scan or quantity.
- Wastage entry with reason and photo.
- Indent to warehouse. Stock count mode.
- Sync health shown on the screen at all times.

Hardware plan

- Thermal printer over USB or network (ESC/POS).
- Barcode scanner as keyboard input.
- Weighing scale: manual entry first, serial or Bluetooth read as a later step.
- Cash drawer kick. UPI QR display on a second screen or on paper.
- Runs on an Android tablet or a modest Windows PC.

POS: offline design and edge cases

How offline works

- Catalog, price versions, store stock snapshot and recent customers are cached on the device.
- Each bill is saved locally first, with a UUID idempotency key. Drafts survive a power cut.
- Invoice numbers come from a series reserved per device and financial year, so numbering stays continuous and GST compliant while offline.
- A background queue pushes bills when online, with retry and backoff. The server rejects duplicates by key.
- Sales are append only. Corrections are a void plus a new bill, never an edit.
- Device clock is recorded along with server receive time, and large skew is flagged.
- A sync dashboard shows last sync age and queue size per store.

Edge cases covered

- Internet drops mid-bill, and returns mid-bill.
- Two billing counters in one store with separate series.
- Price changed at head office while a store is offline: sale uses the price version the device held, and the difference is flagged.
- Item sells beyond system stock: allowed with a negative stock flag for the manager.
- UPI shows success on the phone but not on the terminal: payment pending state with later match.
- Customer adds items after payment starts: bill locks and a new bill opens.
- Festive peak with long queues: hold, quick keys, keyboard shortcuts.
- Store with weak network for days: local retention with a clear warning when the queue grows.

Where I would be careful

- Cost savings must include hardware, rollout, support and a person who owns uptime. Licence fees are one line of the cost.
- Printers, scanners, scales and payment terminals decide how smooth the first week in a store feels. Pilot before promising.
- E-invoicing and e-way bill thresholds and rules need confirmation with their CA before any claim in the demo.
- Run the new POS next to the old one in 3 to 5 stores for a month and compare day totals before any cutover.

Pilot path

- Phase A: shadow mode in 3 to 5 stores, both POS running, daily totals compared.
- Phase B: 25 stores on the new POS for one festive cycle.
- Phase C: wave rollout by city with a rollback option per store.

1. Control Tower

Demo depth: Deep

A single page for the CEO and regional heads that answers one question each morning: what happened yesterday, and what needs attention today. Every number drills down to store, SKU and bill.

Key screens

- Executive summary with sales, gross margin after wastage, stock value, stock cover, wastage, stockout rate.
- Store league table with filters for city, cluster and format.
- Exceptions inbox: stores not synced, negative stock, cash variance, near-expiry value, price mismatch.
- Drilldown from KPI to store to SKU to bill.
- Warehouse view with stock, ageing and dispatch status.

Core workflows

- Daily 7 am snapshot sent to WhatsApp or email for leadership.
- Exception to owner assignment with status and due date.
- Period compare: week on week, month on month, same festival last year.
- Saved views per role.

Metrics it produces

- Net sales, gross margin, gross margin after wastage.
- Average bill value, units per bill, sales per store per day.
- Stock cover in days, sell-through, stockout days.
- Wastage as percent of sales and of purchase cost.
- Cash variance and sync lag by store.

Edge cases covered

- Stores that sync late show totals as of a timestamp, so the headline never silently understates.
- Currency toggle for INR and AED with a stated conversion date.
- Stores opened or closed mid-period are handled in like-for-like comparisons.
- Festival shifts, such as Ramadan moving by about 11 days a year, are handled in compares.

2. SKU master

Demo depth: Deep

One clean catalog. A SKU is a product at a grade, origin and pack size. The catalog follows the website structure, with six collections plus chocolates and spices, and extends it with the operating fields stores and finance need.

Key screens

- Catalog list with search, category tree and status.
- SKU detail: attributes, packs, tax, shelf life, barcodes, price history.
- Kit builder for gift boxes with a bill of materials.
- Price lists by city, cluster and store.
- Bulk import with validation and error report.

Core workflows

- Create SKU from a product template across pack sizes.
- Approval for new SKUs and price changes.
- Lifecycle: draft, active, restricted, delisted.
- Barcode and label generation.

Metrics it produces

- Active SKUs by category, and long-tail share.
- Price change log and margin impact.
- SKUs with no sale in 60 days by store cluster.

Edge cases covered

- Bulk to pack conversion: 25 kg sack to 250 g and 500 g packs, with yield loss.
- MRP lives on the batch. Old packed stock sells at its printed MRP while new batches carry the new one.
- GST rate and HSN with effective dates. Rates stay configurable and are confirmed with their CA.
- Duplicate and near-duplicate SKU detection.
- Seasonal SKUs that switch on and off by date.

3. Vendors, procurement and inbound logistics

Demo depth: Deep

Many vendors supply many SKUs, and cost moves by month. This module holds the price book, builds the landed cost, tracks every purchase from order to receipt, and gives finance the paperwork trail it needs for tax.

Key screens

- Vendor master with documents, currency, payment terms.
- Price book: vendor by SKU with effective dates, currency, Incoterm, MOQ, lead time.
- Landed cost calculator: price, FX, freight, insurance, duty, clearing, handling, inland transport.
- PO list and detail with approval status.
- Shipment tracker: booked, at port, in customs, cleared, in transit, received.
- GRN screen with batch, expiry, MRP and quality check.
- Vendor scorecard and SKU vendor comparison.

Core workflows

- PO from replenishment suggestions or manually.
- Approval by value band.
- Split shipments across warehouses.
- Three-way match: PO, GRN, vendor invoice.
- Debit note on quality rejection.

Metrics it produces

- Landed cost per kg by vendor and month.
- Lead time and its spread, fill rate, price variance against PO.
- Quality rejection percent by vendor.
- Open commitments and payables ageing.

Edge cases covered

- FX moves between PO date and payment date.
- Duty changes while goods are in transit.
- Short landing in a container and partial receipts.
- Vendor substitution when a SKU is short.
- Advance payments and letters of credit.
- IGST on imports recorded from the bill of entry for credit claims.

4. Inventory and warehouse (OMS and WMS)

Demo depth: Deep

The core of the lean ERP replacement. Stock is an append-only ledger by location and batch, so every kilo has a history. Orders move through allocation, picking, dispatch and receipt with status visible to stores and finance.

Key screens

- Stock by location, batch and expiry. Ageing buckets.
- Location tree: country, city, warehouse or store, zone, rack, bin.
- Transfer and dispatch board with in-transit state.
- Store indent inbox with allocation and picking lists (FEFO).
- Repack job screen: input weight, output packs, yield loss.
- Cycle count sheets and variance approval.
- Bulk and corporate order desk.

Core workflows

- Indent to allocation to pick to dispatch to receive.
- Warehouse to warehouse and store to store transfers.
- Receiving with variance capture.
- Quarantine for damaged or doubtful stock.
- Count plan by ABC class.

Metrics it produces

- Stock value, days of cover, ageing by bucket.
- Fill rate on store indents, dispatch time, in-transit value.
- Count accuracy and shrink by location.
- Repack yield by SKU and operator.

Edge cases covered

- Weight drift from moisture loss during storage.
- Mixed batches and partial batch sales.
- In-transit damage and short receipts.
- Negative stock from sync delay, resolved on sync.
- Cold storage and temperature-sensitive items.
- Gift box kits consuming component stock.

5. Wastage and shrink

Demo depth: Deep

Wastage in food retail is where margin leaks quietly. The module makes loss visible at the point it happens, ties it to cost, and nudges action before stock expires.

Key screens

- Quick wastage entry in store and warehouse with reason and photo.
- Near-expiry list at 60, 30 and 15 days with value at risk.
- Markdown and transfer suggestions.
- Shrink report: book stock against count by location.
- Wastage by store, cluster, SKU and vendor.

Core workflows

- Entry to approval by threshold to ledger posting.
- Alert to action: markdown, move to a faster store, add to a mix pack.
- Link wastage back to the vendor batch for claims.

Metrics it produces

- Wastage percent of sales and of cost.
- Value at risk by expiry bucket.
- Shrink percent by category and location.
- Claim recovery from vendors.

Edge cases covered

- Reason codes: expiry, rancid or moisture, pest, damage, spillage, weighing variance, tasting samples, suspected theft, quality reject.
- Tasting samples tracked as a capped expense.
- Loose item weight loss separated from true loss.
- Reversals with full audit trail.

6. Demand and replenishment

Demo depth: Medium

Forecasts that respect Indian festival and winter seasonality, and replenishment suggestions that respect import lead times. The first version uses transparent methods that the team can read and trust.

Key screens

- Seasonality calendar: Diwali, Eid, Ramadan, Navratri, Raksha Bandhan, Holi, Christmas, winter.
- Forecast view by store, SKU and week with accuracy shown.
- Reorder suggestions with safety stock and lead time.
- Festive pre-build plan for gift boxes.
- Assortment suggestions by store cluster.

Core workflows

- Nightly forecast job.
- Suggested PO and suggested store indent, approved by a human.
- Min and max per location reviewed monthly.

Metrics it produces

- Forecast error (WAPE) and bias.
- Stockout days avoided, excess stock value.
- Inventory turns by category.

Edge cases covered

- New SKUs borrow history from similar SKUs.
- Sales during stockouts are treated as censored demand.
- Lunar calendar festivals shift each year.
- Import lead times of 30 to 60 days drive early ordering.
- Middle East store calendar differs from India.

7. Finance and tax reconciliation

Demo depth: Deep

The aim is to track every rupee from vendor invoice to customer bill, and to give finance a clean close. The module prepares reconciled working data. Filing stays with their finance team and CA.

Key screens

- Daily close dashboard by store: cash, UPI, card against expected.
- Settlement reconciliation against payment gateway and bank files, with fees.
- GST workbench: output tax by HSN, rate and state. Input credit from purchases.
- Purchase register matched against the government portal credit statement upload.
- Payables ageing and payment run proposals.
- Stock valuation and margin by SKU after wastage.
- Journal export for Oracle, Tally or CSV.

Core workflows

- Store day-close to finance review to lock.
- Month close with period lock.
- Vendor invoice to three-way match to payable.
- Inter-state stock transfers with tax invoices.
- E-way bill data prepared for transfers, with API integration through a GSP in phase 2.

Metrics it produces

- Cash variance by store and cashier.
- Unreconciled value and age.
- Tax liability and credit position by GSTIN.
- Days to close, and gross margin by SKU, store and vendor.

Edge cases covered

- Cash short and excess with approval.
- Mixed payment bills and payment fees.
- Credit notes and rounding differences.
- Stores under different state GSTINs.
- Backdated entries blocked after period lock.
- Import taxes and customs duty folded into landed cost.

8. Customers and WhatsApp commerce

Demo depth: Light

The website routes visitors to WhatsApp and phone. This module turns that into a tracked order channel and builds a customer record from phone numbers captured at the billing counter.

Key screens

- Customer 360: history, favorite categories, basket size, last visit.
- Segments by recency, frequency and spend.
- WhatsApp order inbox with cart summary and payment link.
- Corporate gifting enquiry and quote screen.
- Campaign list with consent status.

Core workflows

- Phone captured at POS to customer record.
- WhatsApp order to OMS order to dispatch.
- Quote to approval to order for bulk and gifting.
- Festival and repeat-purchase reminders.

Metrics it produces

- Repeat rate, active customers, loyalty share of sales.
- WhatsApp orders and conversion.
- Corporate quote win rate.

Edge cases covered

- Duplicate customers by phone number.
- Consent and opt-out tracking under the data protection rules.
- Template approval and rate limits on WhatsApp business messaging.
- Gift orders shipped to a different address.

9. HR basic

Demo depth: Light

A basic people module sized for a retail chain with a large store workforce. Payroll is out of scope, and the module exports hours to whichever payroll provider they use.

Key screens

- Org chart: head office, region, city, store.
- Employee profile and onboarding checklist.
- Punch in and punch out with store location check.
- Shift roster for stores.
- Leave and work-from-home for head office staff.
- Birthdays and work anniversaries feed.
- Job postings and a basic applicant list that feeds the careers page.

Core workflows

- Offer to onboarding tasks to day one setup.
- Missed punch regularization with manager approval.
- Leave request to approval to balance.
- Requisition to job post to applicant stage.

Metrics it produces

- Attendance and hours by store.
- Overtime, leave balance, open positions.
- Time to fill and attrition by region.

Edge cases covered

- Store staff who log in with a phone number only.
- Staff who move between stores.
- State-wise holiday calendars.
- Location checks have limits and are treated as a guide.

10. Platform

Demo depth: Medium

The shared layer that every module depends on. It keeps the system safe, auditable and usable across entities, states and currencies.

Key screens

- User and role management.
- Approval rules by amount and type.
- Audit log viewer.
- Import and export center.
- Entity, GSTIN and currency settings.
- System health: sync status, job runs, errors.

Core workflows

- Phone OTP login for store staff, email login for head office.
- Row-level scope by store and region.
- Notifications by in-app, WhatsApp and email.
- API keys for other systems.

Metrics it produces

- Active users by role.
- Approval turnaround time.
- Job success rate and sync lag.

Edge cases covered

- One person with roles in several stores.
- Multiple legal entities and state registrations.
- INR and AED with time zones.
- Backups and restore drills.
- Every write carries user, device and time.

Core data model

The entities that carry the whole system. Everything else is a view over these.

Entity	Key fields	Note
location	id, type (warehouse, store, in-transit), city, state, country, entity, GSTIN	Tree with parent location for zones and bins.
sku, sku_pack	product, grade, origin, pack size, HSN, shelf life, uom, barcode	Pack conversion rules for repacking.
vendor, vendor_sku_cost	vendor, sku, currency, incoterm, moq, lead time, price, effective_from, effective_to	Cost history is never overwritten.
purchase_order, po_line, shipment	vendor, status, expected dates, tracking stages, charges	Shipment holds freight, duty and clearing.
grn, grn_line, batch	po ref, batch no, expiry, MRP, accepted and rejected qty	Batch is the unit of traceability.
stock_ledger	location, batch, qty, movement type, document ref, timestamp, user	Append only. Stock on hand is a sum.
transfer, repack_job	from, to, status, input and output batches, yield loss	In-transit is a location.
sale, sale_line, payment	store, device, invoice no, idempotency key, business date, tax split	Immutable once synced.
day_close, settlement	store, expected and counted cash, gateway files, variance, lock status	Feeds finance reconciliation.
wastage_entry	location, batch, reason, qty, cost, approver, photo	Posts to the ledger on approval.
customer, consent	phone, segments, loyalty points, consent flags	Phone is the identity key.
employee, attendance, roster	role, store, punches, shift, leave	Light version in the demo.
journal_entry	date, account, debit, credit, source document	Export format for any ERP.

Demo data plan

The demo has to look like their business, so the data is built to behave like it

What gets generated

- 500 stores across major Indian cities, with a small group of Middle East stores to show multi-currency.
- 20 warehouses, with a regional hub pattern.
- 1,200 SKUs built from the public collections, with names and pack sizes that match what stores sell. Prices and costs are modeled, and the demo says so.
- About 60 vendors, a mix of import and domestic, with several vendors per SKU and cost that moves by month.
- 24 months of sales with weekday patterns, winter uplift, Diwali, Eid and Ramadan peaks, and gift box spikes.
- Batches with expiry, wastage, transfers, repacking, store indents, day closes and settlements.

Volume handling

- Full-scale history is generated locally into Parquet and DuckDB.
- The hosted demo carries a trimmed slice: line-level bills for a subset of live stores, and daily aggregates for the rest.
- The final split depends on the hosting tier and is settled on day 1.

Findings planted in the data

Each one gives the CEO a concrete story in under a minute.

- One warehouse with shrink well above its peers.
- One vendor whose landed cost rose for three months while a cheaper vendor stayed available.
- A cluster of stores with chronic stockouts on fast-moving almonds and cashews.
- A store with repeated cash shortage on weekends.
- Three stores not syncing for two days.
- A fresh dried fruit SKU with heavy expiry loss in smaller stores.
- Gift box component shortage ahead of a festival.
- A month where GST input credit in the books differs from the portal statement.

Labeling

The app shows a visible demo data label. Nothing in it claims to be their real numbers.

The 10 minute demo story

Built for a CEO. Short, concrete, and ending on ownership and cost.

Minute	Scene	What the CEO sees
0 to 1	Control Tower	Yesterday across 500 stores. Eight exceptions already flagged.
1 to 3	Drill into a problem	Weekend cash shortage in one store, traced to the cashier and shift. Shrink hotspot in one warehouse.
3 to 5	POS on a tablet	Bill loose almonds by weight. Switch off Wi-Fi. Keep billing. Switch on. Watch the queue sync and the sale appear in the dashboard.
5 to 6	Procurement	Same SKU from three vendors, landed cost compared, and the PO suggestion it produces.
6 to 7	Inventory and wastage	Near-expiry stock value at risk and the suggested move to a faster store.
7 to 8	Finance	Month close view, settlement reconciliation, GST working data matched against the portal statement.
8 to 9	Customers and WhatsApp	A WhatsApp order flowing into dispatch.
9 to 10	Ownership	Roadmap, pilot path, and what a first 90 days looks like.

Modules 6, 8, 9 and 10 appear as short moments, since the story is margin, control and speed of finance.

One week build plan

Close to seven days. The plan is tight and the cut list is decided in advance.

Day	Work	Output
1	Lock scope. Collect catalog and brand assets from the site. Pick palette. Finalize schema and data volume plan.	Schema, design tokens, catalog seed
2	Data generator. SKU master. Vendors, price book, landed cost.	Seeded database with 24 months of history
3	Inventory ledger, GRN, transfers, repack, indents.	Warehouse flows working end to end
4	POS PWA with offline storage and sync. Day close.	Billing demo with network drop and sync
5	Wastage, finance reconciliation, Control Tower.	CEO story running on live data
6	Forecast and replenishment. Light modules. Roles and audit.	Remaining screens
7	QA, performance, rehearsal, final deck and PDF export.	Demo script and backup recording

Cut list if time runs out, in this order

- HR basic and Customers: designed screens with seeded data, no workflow.
- Demand and replenishment: suggestions from a simple seasonal average, without accuracy views.
- Platform: roles and audit only.
- Never cut: POS offline demo, Control Tower, procurement landed cost, wastage, finance close.

Risks

- A one-person week with this scope leaves little room for rework. A recorded backup of the demo is part of the plan.
- Hosted data size and cold starts can slow the demo. Paid tiers for the week, and a local fallback.
- Real catalog and brand colors from the site help the look, and tax rules in the demo are labeled as configurable and subject to their CA.

Business case frame and open questions

Numbers to find out in the room, and questions that change the build

Number to find out	Why it matters
POS licence, support and hardware cost per store per year	Sets the savings line honestly, including what the new setup would cost.
Finance hours per month spent on reconciliation, and days to close	Shows the value of automated reconciliation.
Wastage and shrink as percent of sales, if known	Gives the size of the prize for module 5.
Stockout days on top SKUs	Shows the value of replenishment.
Number of legal entities and GSTINs	Shapes finance and invoicing design.
Oracle quote, scope and timeline	Frames the lean alternative against a real number.

Open questions

- The website carries a return policy page. Is there a returns flow in practice, even a small one?
- Can the current POS vendor export data nightly, even as CSV?
- Do stores sell loose items by weight, and on which scales?
- Which payment terminals and UPI setups are in stores?
- How are labels and barcodes printed today?
- Which Middle East entities exist, and in which currencies?
- Is anything in the current stack staying, such as Tally or a payroll provider?

Design system foundations

Shared rules across all four palette options

Typography

- Headings: a serif or a clean sans, depending on palette option. Body and UI: Inter. Numbers and tables: tabular figures.
- Scale: 12, 14 (base), 16, 20, 24, 32. Line height 1.4 for text, 1.2 for headings.
- POS uses a larger base of 16 and numerals at 28 to 40 for totals.

Spacing and shape

- 4 point grid. Steps 4, 8, 12, 16, 24, 32.
- Radius 8 for cards and inputs, 999 for chips.
- Light borders over heavy shadows. One elevation level for menus and dialogs.

Density modes

- POS mode: 48 px minimum touch targets, high contrast, keyboard shortcuts.
- Back office mode: compact tables, 32 px rows, sticky headers and filters.

Components

- App shell with sidebar, top bar, role-based navigation.
- KPI card with value, delta and sparkline.
- Data table with sort, filter, column picker, export.
- Filters bar: date range, city, cluster, category.
- Forms with inline validation and unit-aware inputs (kg, g, packs).
- Status chips, toasts, empty states, approval dialogs.
- Sync indicator for the POS: online, offline, syncing, queue count.

Charts

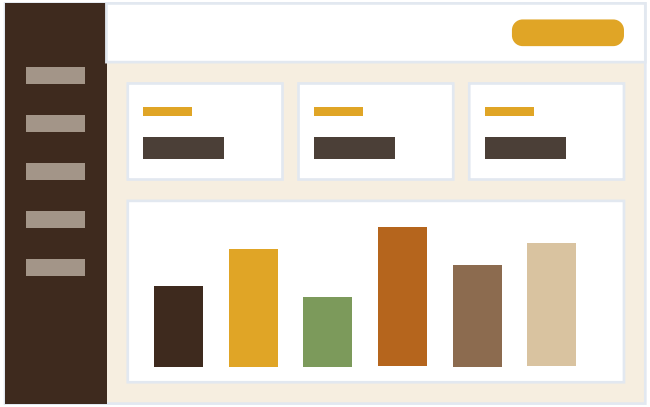
- Six series colors per palette, tested for color blind separation.
- Direct labels where possible, legends only when needed.
- Good and bad use the status colors, never the series colors.

Accessibility

- Text contrast of 4.5 to 1 or higher. Never rely on color alone for status.
- Full keyboard support in the POS and in tables.

Palette options, set 1

Warm and brand-flavored directions



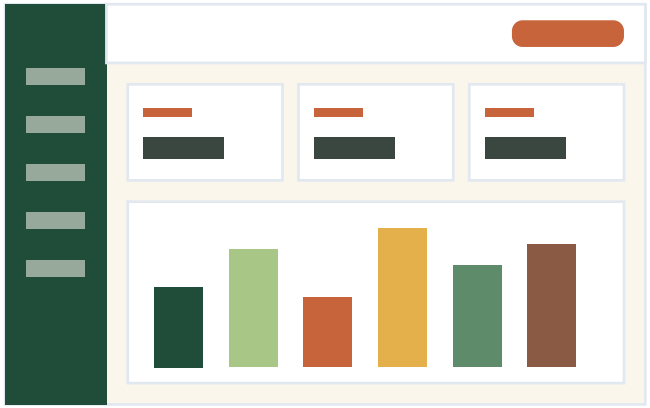
Option A: Walnut and Saffron

Warm browns, cream and saffron. Reads like the product itself.

Primary	Secondary	Accent	Page bg	Surface	Text	Success	Warning	Danger
#3E2A1E	#7C9A5B	#E0A526	#F6EEE0	#FFFFFF	#2B1D14	#4F7A3A	#C98A0B	#B23A2E

Chart 1	Chart 2	Chart 3	Chart 4	Chart 5	Chart 6
#3E2A1E	#E0A526	#7C9A5B	#B5651D	#8C6B4F	#D9C3A0

Type: Fraunces for headings, Inter for UI
Good for: Decks, POS receipts, the customer-facing side. Charts need care with the browns.



Option B: Pistachio Fresh

Deep forest green with pistachio and a terracotta accent.

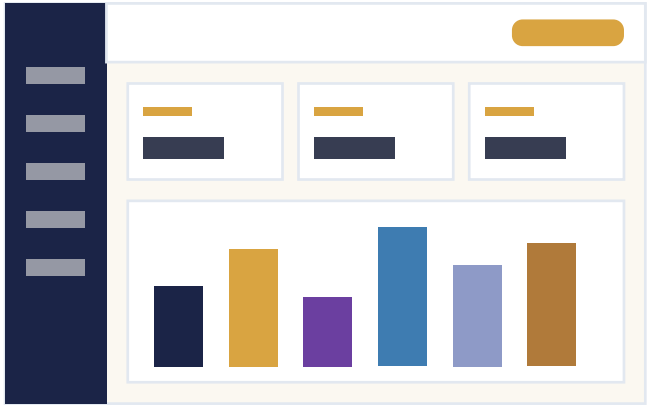
Primary	Secondary	Accent	Page bg	Surface	Text	Success	Warning	Danger
#1F4D3A	#A8C686	#C8643B	#FAF6EC	#FFFFFF	#17261F	#2F7D4F	#D9922B	#B23A2E

Chart 1	Chart 2	Chart 3	Chart 4	Chart 5	Chart 6
#1F4D3A	#A8C686	#C8643B	#E3B04B	#5E8C6A	#8A5A44

Type: Lora for headings, Inter for UI
Good for: A fresh, natural identity. Strong for stores and marketing screens.

Palette options, set 2

Premium and operational directions



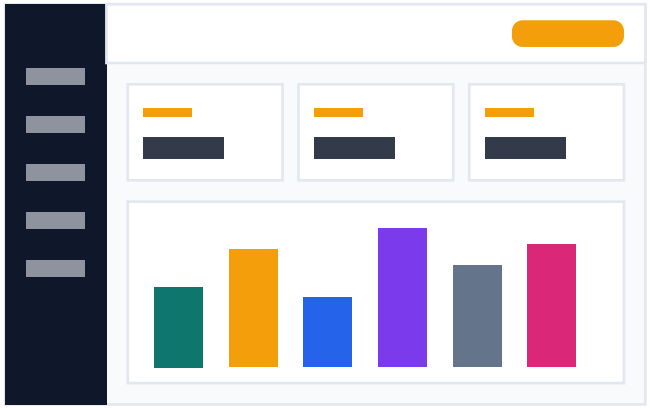
Option C: Royal Indigo

Indigo with gold. Premium and gifting-led.

Primary	Secondary	Accent	Page bg	Surface	Text	Success	Warning	Danger
#1B2447	#6B3FA0	#D9A441	#FBF8F1	#FFFFFF	#141A33	#2E7D5B	#D9A441	#B3342F

Chart 1	Chart 2	Chart 3	Chart 4	Chart 5	Chart 6
#1B2447	#D9A441	#6B3FA0	#3E7CB1	#8E9AC7	#B07A3A

Type: Playfair Display for headings, Inter for UI
Good for: Gift boxes and corporate gifting. Less suited to dense tables.



Option D: Clean Ops

Slate, teal and amber on a near-white base.

Primary	Secondary	Accent	Page bg	Surface	Text	Success	Warning	Danger
#0F172A	#0F766E	#F59E0B	#F8FAFC	#FFFFFF	#0F172A	#16A34A	#F59E0B	#DC2626

Chart 1	Chart 2	Chart 3	Chart 4	Chart 5	Chart 6
#0F766E	#F59E0B	#2563EB	#7C3AED	#64748B	#DB2777

Type: Inter for headings and UI, JetBrains Mono for figures
Good for: Dashboards, tables and long working sessions. Easiest charts.

Recommendation and next steps

Recommendation

Use Option D as the base for the back office and the POS, and bring in one warm accent from their actual brand once I have the logo values. Dense tables and charts read better on a neutral surface, and the brand shows up through the accent, the logo and the receipt. Options A and B suit the deck cover and customer-facing pages. Option C fits a dedicated gifting page.

Brand colors

I could not read the logo colors from the site text. Send a screenshot of the logo or the site header, or I can pick them from the logo file, and the tokens above will be updated in one place.

Next steps

- You confirm the palette direction and the cut list.
- I collect the catalog and brand assets and generate the seed data.
- Build in the order of the week plan, with a recorded backup of the demo.
- Final deck and demo script on day 7.